31 August 1970
SUPERSEDING
MIL-B-43254A
18 February 1969

MILITARY SPECIFICATION

BRUSH, ANIMAL GROOMING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope.- This specification covers two types of animal grooming brushes.

1.2 Classification.- The brushes covered by this specification shall be of the following types, as specified (see 6.2).

Type I - Small
Type II - Large

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

H-S-951	-	Synthetic Filaments for Brushes; General Specification For.
L-P-394	-	Plastic Molding Material, (Propylene Plastics, Injection and Extrusion).
KK-L-156	-	Leather, Cattlehide, Harness, Vegetable-Tanned.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.

MILITARY

MIL-P-116	-	Preservation, Methods Of.

FSC 3770

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Specification D785-65 - Method of Test for Rockwell Hardness of Plastics and Electrical Insulating Materials
Specification D2016-65 - Methods of Test for Moisture Content of Wood

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

* 3.1 First article.- This specification contains provisions for first article approval (see 4.2, 6.2, and 6.3).

3.2 Materials.-

3.2.1 Wood.- The wood shall be close grained hardwood such as beech, birch, or maple and shall be thoroughly and uniformly dried to a moisture content at time of fabrication, not exceeding 14 percent (see 4.3.1.1).

3.2.2 Polypropylene.- The polypropylene shall be virgin, general purpose, homopolymer polypropylene meeting the requirements of type I of L-P-394, except the property values shall be as specified in table I, when tested in accordance with 4.3.1.2. Clean, unburned plastic material in the form of imperfect parts, sprues, runners, or other scrap of the same composition as the virgin material and produced in molding or finishing operations, may be reground and mixed with the virgin material.

TABLE I.- Polypropylene property values

Property	Units		Property values at 23°C. (73°F.)
Density	Grams per cubic centimeter	Min.	0.901
		Max.	0.907
Tensile yield	Pounds per square inch	Min.	4900
Izod impact strength	Foot-pounds per inch	Min.	0.5
Rockwell hardness	R-scale	Min.	96
		Max.	104

* 3.2.3 Filler material, styrene.- The styrene plastic filler material shall be 0.016 inch diameter and shall conform to class S, type III of H-S-951.

* 3.2.4 Filler material, Palmyra.- The Palmyra filler material shall be pure, stiff Malibar, Tuticorin, or Coconada, Palmyra fiber, of medium length (12 inch - 14 inch) fibers, dressed and cut to length to meet all requirements of table II.

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* 3.2.5 Staple wire.- Staple wire shall be not less than 0.023 inch diameter commercial steel staple wire, treated to resist corrosion in accordance with commercial practice.

3.2.6 Leather.- Leather for the strap shall be russet color, conforming to type I or II, class 2, selection A, B, or C, extra light weight of KK-L-156.

* 3.2.7 Escutcheon pins.- Escutcheon pins shall be 1/16-inch diameter by 3/4-inch long with 3/8-inch diameter head. Pins shall be corrosion resistant metal, or steel treated to resist corrosion.

* 3.3 Construction.- The brushes shall be constructed similar to figures 1 and 2 and shall conform to the requirements specified herein and table II. Figures 1 and 2 are intended for illustration purposes only and are not intended to preclude brushes otherwise meeting requirements of this specification. The type I brushes shall have the filler material specified in 3.2.3. The type II brushes shall have the filler material specified in 3.2.4.

TABLE II.- Physical requirements per unit

Block:	Type I	Type II
Length	4-7/8 - 5 inches	8 - 8-1/4 inches
Width	2-5/8 - 2-3/4 inches	4 - 4-1/8 inches
Thickness, at center	7/8 - 1 inch	3/4 - 13/16 inch
Thickness, at ends	1/2 - 9/16 inch	3/4 - 13/16 inch
Holes:		
Quantity	106	91
Diameter	0.116 - 0.118 inch	0.264 - 0.268 inch
Depth	1/4 - 5/16 inch	5/16 - 3/8 inch
Filler material:		
Type	Styrene plastic	Palmyra fiber
Weight	0.450 - 0.500 ounces	3.06 - 3.10 ounces
Length, clear of block	3/4 - 13/16 inch	1-5/16 - 1-3/8 inch
Strap:		
Length	6-7/8 - 7 inches	7-5/8 - 7-3/4 inches
Width	5/8 - 11/16 inch	1-5/8 - 1-11/16 inches
Thickness	7/64 - 8/64 inch	10/64 - 11/64 inch

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3.3.1 Block.- The block shall be fabricated from either the wood specified in 3.2.1 or the polypropylene specified in 3.2.2. The block shall be oval shaped, made in one piece with all edges rounded. The specified number of holes for the tufts shall be equally spaced. The polypropylene block shall have two molded round tips located at opposite sides of the width, centered along the length of the block, 1/4-inch from back surface, for strap attachment.

3.3.2 Staple set.- The filler material specified in 3.2.3 or 3.2.4 as applicable, shall be cut to double length and center-folded in quantities to insure individual tufts that fit snugly when inserted in the holes provided in the block. Stapling shall be accomplished so the staples and filler material are driven to the bottom of the holes, with the staple securely imbedded into the wood or polypropylene. Staples shall be made of wire specified in 3.2.5. Filler material shall be evenly trimmed to required length.

3.3.3 Retention of filler material.- The filler material shall be so secured in the block that, when tested as specified in 4.4.1, it shall not be possible to remove the filler material from the block without breaking the filler material.

* 3.3.4 Hand strap.- The hand strap shall be made from leather specified in 3.2.6. The strap for the type I brush shall be attached to the block, with the flesh side toward the block, by two escutcheon pins conforming to 3.2.7. The strap for the type II brush shall be attached in the same manner except four pins shall be used. The type II strap shall be centered on the length of the block with the ends of the strap even with the bottom of the block and the pins shall penetrate the strap 1/4 inch in from each side and 1/4 inch up from each end of the strap. For attachment of strap to the polypropylene block, each end of the strap shall be slit for attaching to the two molded tips.

3.4 Finish.- Wood blocks shall be smoothly sanded or tumbled smooth and given a clear organic coating.

3.5 Marking for identification.- The supplier shall permanently and legibly indent his name, or trademark of such known character that it is identifiable with said manufacturer, on the block of each brush.

3.6 Workmanship.- The brushes shall have no exposed sharp or rough edges, or splinters. Escutcheon pins and staples shall be driven tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his

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own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table III, dimensions specified in 4.3.2.2, and tested in accordance with 4.4.1. The presence of any visual defect, any dimension not within specified requirements, or failure to comply with test requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Moisture content of wood blocks.- When wood blocks are furnished, the moisture content shall be determined in accordance with ASTM D 2016-65 at time of assembly. The lot shall be all wood blocks prepared under the same conditions, stored as a group, and offered for inspection at one time. The sample unit shall be one block. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units. When testing is performed by the electric moisture meter method, 3 determinations shall be made on each sample unit. The average of the 3 determinations shall be the moisture content of the sample unit.

4.3.1.2 Property values of polypropylene.- Polypropylene shall be tested for density, tensile yield strength, and Izod impact strength in accordance with L-P-394. Testing for Rockwell hardness shall be in accordance with procedure A of ASTM D 785-65. Failure to meet specification requirements shall be cause for rejection of the polypropylene.

* 4.3.2 End product inspection.- The lot shall consist of all completely fabricated brushes of one type, offered for inspection at one time. The sample unit shall be one brush.

4.3.2.1 Visual examination.- The completely fabricated brush shall be examined for defects listed in table III. The inspection level shall be level II and the AQL shall be 2.5, expressed in terms of defects per hundred units.

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* TABLE III.- Classification of defects for visual characteristics

Examine	Defect
Finish	Not sanded or tumbled smooth (wood block only).
Block	Not finished as specified. Not oval shaped, edges not rounded. Splintered or loose grain (wood), exposed staple point, or sharp edge. Pin or tip missing, or loose.
Filler material	Not clean, evidence of dirt or foreign matter. Not evenly distributed in brush head. Not evenly trimmed, loose tufts.
Hand strap	Missing, not properly attached.
Marking	Omitted, incomplete, illegible.

4.3.2.2 Dimensional examination.- Examination shall be made of the brush for compliance with dimensions specified. Any dimensions not within the specifiefd requirements shall be classed as a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.3.2.3 End item testing.- Filler material retention shall be tested as specified in 4.4.1. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units. There shall be ten determinations per sample unit and the results shall be reported as pass or fail.

4.3.3 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing, and marking requirements are in compliance with section 5. Defects shall be scored in accordance with table IV. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 2.5 expressed in terms of defects per hundred units.

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* TABLE IV.- Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Component missing, damaged, or defective.
Workmanship	Inadequate application of components such as loose strapping or tape banding, or bulging or distortion of contents. Intermediate package contains more or less than required.
Weight	Weight exceeds requirements.

* 4.3.4 Examination of shipping container.- When shipping containers are required to be in accordance with PPP-B-601 or PPP-B-636, examination for defects in closure and strappping shall be made in accordance with the appendix of the applicable specification.

4.4 Tests.-

* 4.4.1 Filler material retention.- In order to determine compliance with the requirement specified in 3.3.3, individual strands of filler material shall be subjected to a pull test of 4 lbs. minimum for type I brush and 27 lbs. for type II brush.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Unit packaging.- Each brush shall be packaged in a snug-fitting paperboard box or sleeve in accordance with Method III of MIL-P-116. The box or sleeve shall be secured in a manner that will assure containment of the brush within the package.

5.1.1.2 Intermediate packaging.- Brushes, unit packaged as specified in 5.1.1.1, shall be packaged in multiples of five in a snug-fitting intermediate fiberboard box conforming to class weather-resistant, grade W5c or W5s, style RSC of PPP-B-636. Each box shall be closed in accordance with the appendix of the box specification.

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5.1.2 Level C.- Brushes shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.2).

* 5.2.1 Level A.- Brushes of one type only, packaged as specified in 5.1 shall be packed in a shipping container conforming to overseas type, style A of PPP-B-601. Each shipping container shall be closed and reinforced in accordance with the appendix of the box specification. The gross weight of each shipping container shall not exceed 125 pounds.

* 5.2.2 Level B.- Brushes of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting container conforming to type CF (variety SW) or SF, class domestic, grade 275, style RSC of PPP-B-636. Each container shall be closed in accordance with Method II of the appendix of the PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

* 5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

* 5.2.3 Level C.- Brushes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages, intermediate packages, shipping containers, and palletized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use.- The brushes covered by this specification are intended for use in the grooming of Military animals, such as sentry and scout dogs.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type brush required (see 1.2).

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(c) First article (see 3.1, 4.2, and 6.3).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant boxes are required for level B shipments (see 5.2.2.1).

* 6.3 First article.- The item covered by this specification requires first article approval under the appropriate provisions of paragraph 7-104.55 of the Armed Service Procurement Regulations. The first article should be a preproduction sample and should consist of one unit. The contracting officer should include specific instructions, in all procurement instruments, regarding arrangements for examination, test, and approval of the first article.

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 3770-0100

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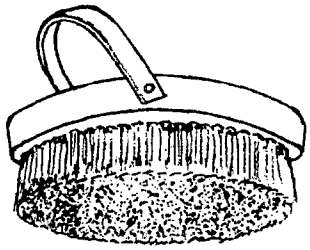


FIGURE 1 - TYPE I (SMALL)

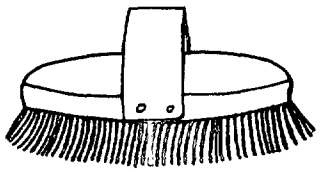


FIGURE 2 - TYPE II (LARGE)

BRUSH, ANIMAL GROOMING

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Brush, Animal Grooming MIL-B-43254B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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